

Power Supply Systems

Part II

Phase 4: Distributor Fed at One End

Aim: To learn distributor fed at one end from absolute zero and become ready for semester numericals.

Style: Simple English, SN ma'am style, diagram first, current calculation next, voltage drop last.

Target: Current-loading diagram, voltage-drop diagram, concentrated loading, uniformly distributed loading, mixed loading and minimum voltage.

Subject: Power Supply Systems

Section: Part II

Phase: 4 – Distributor Fed at One End

Level: Absolute zero to semester-question-ready

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1 Phase 4 Roadmap

Phase 4 Goal

In Phase 1, we learnt that a distributor supplies consumers through tapplings. Now we learn how to calculate current, voltage drop and voltage at each load point when the distributor is fed from one end.

Distributor → Loads tapped at points → Section currents → Voltage drops → Voltage at each load point

Memory Hook

Distributor numericals are not random. Every problem follows one rule:

First find current in each section, then find voltage drop in each section.

2 Meaning of Distributor Fed at One End

Definition

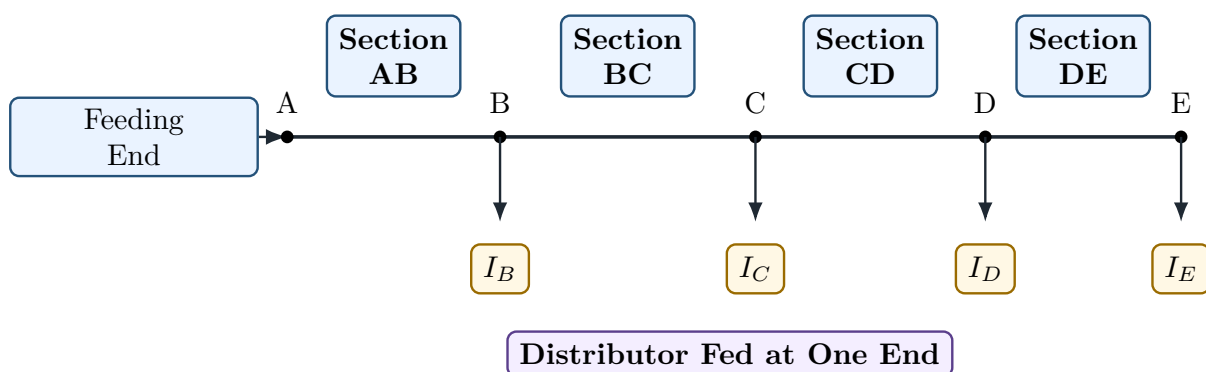
A **distributor fed at one end** is a distributor in which supply is given only from one end and all loads are supplied along the length of the distributor.

In this system, current is maximum near the feeding point and gradually decreases as loads are tapped off.

Current is maximum at feeding end and minimum at far end.

Voltage is maximum at feeding end and minimum at far end.

3 Basic Diagram of Distributor Fed at One End



How to Read the Diagram

Word/Symbol	Meaning
Feeding End	The end where supply voltage is applied. Here it is point A side.
A, B, C, D, E	Points on the distributor. Loads are tapped at B, C, D and E.
Section AB	Part of distributor between A and B. Current in this section supplies all loads after A.
Section BC	Part between B and C. Current in this section supplies loads at C, D and E.
I_B, I_C, I_D, I_E	Load currents taken by consumers at points B, C, D and E.
Downward arrows	These show current being tapped from the distributor by consumers.
Distributor Fed at One End	Supply enters only from one side, so current flows from A towards E.

4 Most Important Rule: Section Current

The current in any section is equal to the sum of all load currents beyond that section.

For the above diagram:

$$I_{AB} = I_B + I_C + I_D + I_E$$

$$I_{BC} = I_C + I_D + I_E$$

$$I_{CD} = I_D + I_E$$

$$I_{DE} = I_E$$

Memory Hook

In a distributor fed at one end:

Section current = sum of loads to the right of that section

5 Voltage Drop in Each Section

Voltage drop in a conductor section is:

$$\Delta V = IR$$

For different sections:

$$\Delta V_{AB} = I_{AB}R_{AB}$$

$$\Delta V_{BC} = I_{BC}R_{BC}$$

$$\Delta V_{CD} = I_{CD}R_{CD}$$

$$\Delta V_{DE} = I_{DE}R_{DE}$$

Total voltage drop up to point E is:

$$\Delta V_{AE} = I_{AB}R_{AB} + I_{BC}R_{BC} + I_{CD}R_{CD} + I_{DE}R_{DE}$$

If voltage at A is V_A , then:

$$V_B = V_A - \Delta V_{AB}$$

$$V_C = V_B - \Delta V_{BC}$$

$$V_D = V_C - \Delta V_{CD}$$

$$V_E = V_D - \Delta V_{DE}$$

Common Mistake

Always check whether resistance is given **per conductor** or **for both conductors**. For a 2-wire DC distributor, if resistance is given per conductor, loop resistance is double.

6 Resistance Convention in DC Distributor

In DC distributor numericals, two different conventions may appear.

6.1 Case 1: Resistance Given for Both Conductors

If the problem says:

resistance of distributor is $r\Omega$ for both conductors

then use directly:

$$\Delta V = Ir$$

6.2 Case 2: Resistance Given Per Conductor

If the problem says:

resistance is $r\Omega$ per conductor

then the current goes through outgoing and return conductor. So total loop resistance is:

$$R_{\text{loop}} = 2r$$

and voltage drop is:

$$\Delta V = I(2r)$$

Semester Answer Style

In semester solution, write one line clearly:

Assume given resistance is loop resistance unless stated per conductor.

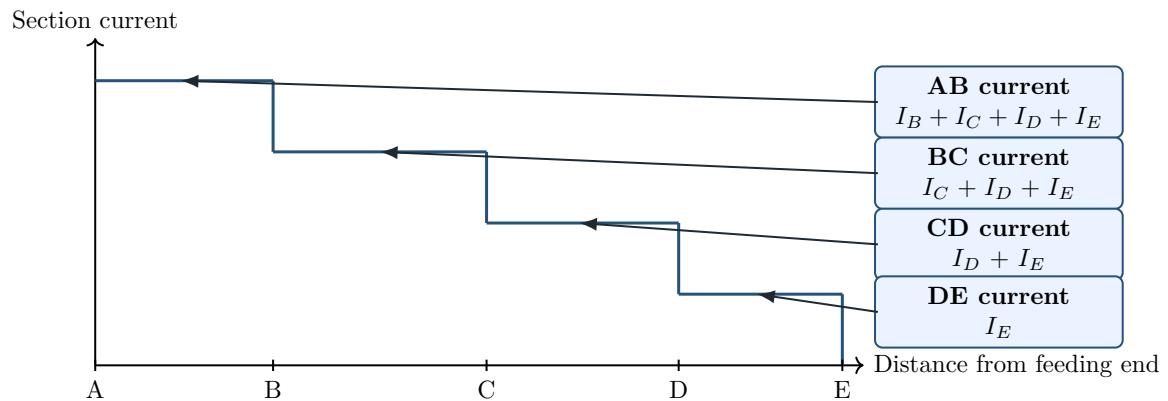
or

Since resistance is per conductor, loop resistance is doubled.

7 Current-Loading Diagram

Definition

A **current-loading diagram** shows how current changes along different sections of the distributor.



How to Read the Diagram

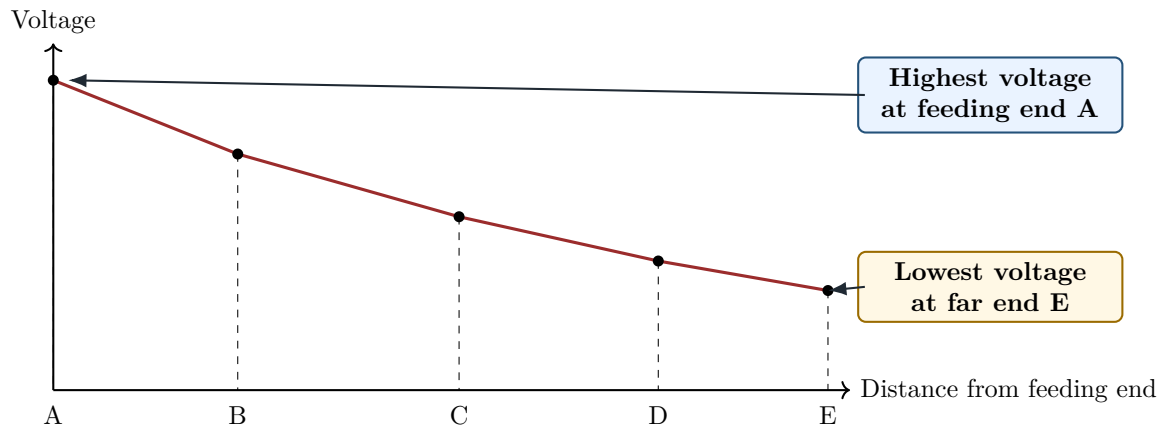
Word/Symbol	Meaning
Distance from feeding end	Horizontal axis. It shows movement from A towards E.
Section current	Vertical axis. It shows current flowing in each section.
A, B, C, D, E	Load points on the distributor.
AB current	Current in section AB. It is maximum because it supplies all loads.
BC current	Current in section BC. It is less than AB because load at B has already been tapped.
CD current	Current in section CD. It supplies only D and E loads.
DE current	Current in section DE. It supplies only the far-end load E.
Step shape	Current suddenly decreases at each load point because some current is taken by the load.

8 Voltage Profile Diagram

Definition

A **voltage profile diagram** shows how voltage decreases along the length of a distributor.

For concentrated loads, voltage falls section by section. The slope is more where current is more.



How to Read the Diagram

Word/Symbol	Meaning
Voltage	Vertical axis. It shows voltage available at each point of the distributor.
Distance from feeding end	Horizontal axis. It shows distance from supply point A.
Point A	Feeding point. Voltage is highest here.
Points B, C, D	Intermediate load points. Voltage decreases as we move from A to E.
Point E	Farthest point. In one-end feeding, voltage is usually minimum here.
Sloping line	Voltage drop along the distributor. Greater slope means greater current and more voltage drop per metre.
Highest voltage	Supply voltage at feeding end.
Lowest voltage	Minimum consumer voltage, generally at farthest end.

9 Case 1: Concentrated Loads

Definition

Concentrated loads are loads connected at specific points of the distributor.

For example, loads may be connected at B, C, D and E only.

9.1 General Formula

For a distributor A–B–C–D–E:

$$\Delta V_{AE} = I_{AB}R_{AB} + I_{BC}R_{BC} + I_{CD}R_{CD} + I_{DE}R_{DE}$$

where,

$$I_{AB} = I_B + I_C + I_D + I_E$$

$$I_{BC} = I_C + I_D + I_E$$

$$I_{CD} = I_D + I_E$$

$$I_{DE} = I_E$$

Semester Answer Style

For concentrated-load numerical, make a table:

Section	Current	Resistance	Voltage drop
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10 Solved Example 1: Concentrated Loads

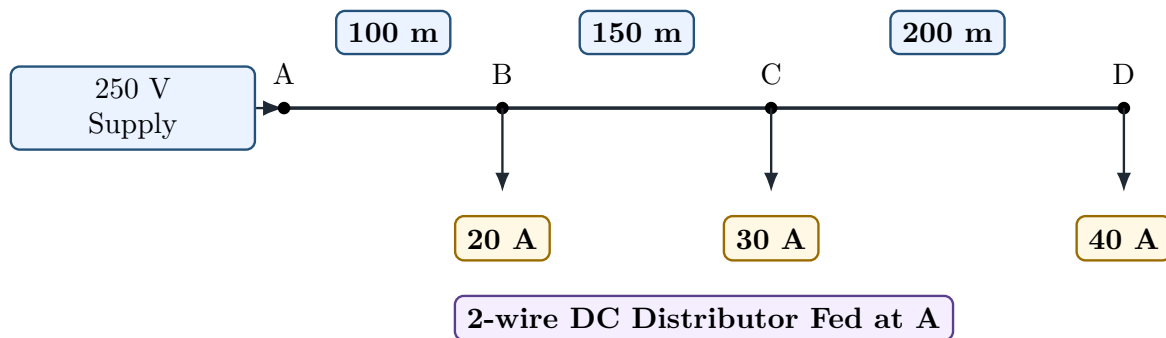
10.1 Problem

A 2-wire DC distributor AB is fed at end A at 250 V. Loads of 20 A, 30 A and 40 A are tapped at points B, C and D respectively. Distances are:

$$AB = 100 \text{ m}, \quad BC = 150 \text{ m}, \quad CD = 200 \text{ m}$$

Resistance of distributor is $0.0005\Omega/\text{m}$ for both conductors. Find voltages at B, C and D.

10.2 Line Diagram



How to Read the Diagram

Word/Symbol	Meaning
250 V Supply	Voltage applied at feeding point A.
A	Feeding end. Calculation starts from this point.
B, C, D	Load tapping points. Consumers take current from these points.
20 A, 30 A, 40 A	Load currents at B, C and D.
100 m, 150 m, 200 m	Lengths of sections AB, BC and CD.
2-wire DC Distributor	Distributor has outgoing and return conductors. Given resistance is already for both conductors.

10.3 Step 1: Section Currents

$$I_{AB} = 20 + 30 + 40 = 90 \text{ A}$$

$$I_{BC} = 30 + 40 = 70 \text{ A}$$

$$I_{CD} = 40 \text{ A}$$

10.4 Step 2: Section Resistances

Given resistance:

$$r = 0.0005\Omega/\text{m}$$

For section AB:

$$R_{AB} = 100 \times 0.0005 = 0.05\Omega$$

For section BC:

$$R_{BC} = 150 \times 0.0005 = 0.075\Omega$$

For section CD:

$$R_{CD} = 200 \times 0.0005 = 0.10\Omega$$

10.5 Step 3: Voltage Drops

$$\Delta V_{AB} = I_{AB} R_{AB} = 90 \times 0.05 = 4.5 \text{ V}$$

$$\Delta V_{BC} = I_{BC} R_{BC} = 70 \times 0.075 = 5.25 \text{ V}$$

$$\Delta V_{CD} = I_{CD} R_{CD} = 40 \times 0.10 = 4.0 \text{ V}$$

10.6 Step 4: Voltages at Load Points

$$V_B = 250 - 4.5 = 245.5 \text{ V}$$

$$V_C = 245.5 - 5.25 = 240.25 \text{ V}$$

$$V_D = 240.25 - 4.0 = 236.25 \text{ V}$$

$$V_B = 245.5 \text{ V}$$

$$V_C = 240.25 \text{ V}$$

$$V_D = 236.25 \text{ V}$$

Semester Answer Style

Final sentence: The minimum voltage occurs at the farthest point D and is 236.25 V.

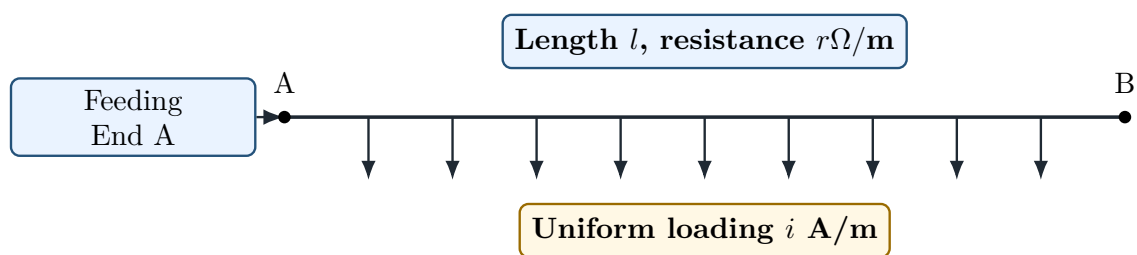
11 Case 2: Uniformly Distributed Load

Definition

A **uniformly distributed load** means load is spread uniformly along the whole length of the distributor.

If loading is i A/m, then every metre length takes i ampere.

11.1 Uniform Loading Diagram



How to Read the Diagram

Word/Symbol	Meaning
Feeding End A	Supply is applied at point A only.
A and B	A is the feeding end and B is the far end.
Uniform loading i A/m	Every metre of the distributor takes the same current i .
Downward arrows	Continuous load spread along the distributor. Arrows represent many small loads.
Length l	Total length of distributor AB.
Resistance $r\Omega/\text{m}$	Resistance per metre of distributor. It may be loop resistance depending on problem statement.

12 Derivation: Voltage Drop for Uniformly Loaded Distributor

Let:

l = length of distributor

i = load current per metre

r = resistance per metre

Take a small section of length dx at a distance x from the feeding end.

Current flowing through this section is the load beyond that point:

$$I_x = i(l - x)$$

Resistance of small section:

$$dR = rdx$$

Small voltage drop:

$$dV = I_x dR$$

$$dV = i(l - x)r dx$$

Total voltage drop from A to B:

$$\Delta V = \int_0^l i(l - x)r dx$$

$$\Delta V = ir \int_0^l (l - x) dx$$

$$\Delta V = ir \left[lx - \frac{x^2}{2} \right]_0^l$$

$$\Delta V = ir \left(l^2 - \frac{l^2}{2} \right)$$

$$\Delta V = \frac{ir l^2}{2}$$

If total current is:

$$I = il$$

and total resistance is:

$$R = rl$$

then:

$$\Delta V = \frac{IR}{2}$$

Memory Hook

For uniformly loaded distributor fed at one end:

$$\text{Voltage drop} = \frac{1}{2} \times \text{total current} \times \text{total resistance}$$

13 Current and Voltage Profile for Uniform Loading

13.1 Current Profile

For uniform loading:

$$I_x = i(l - x)$$

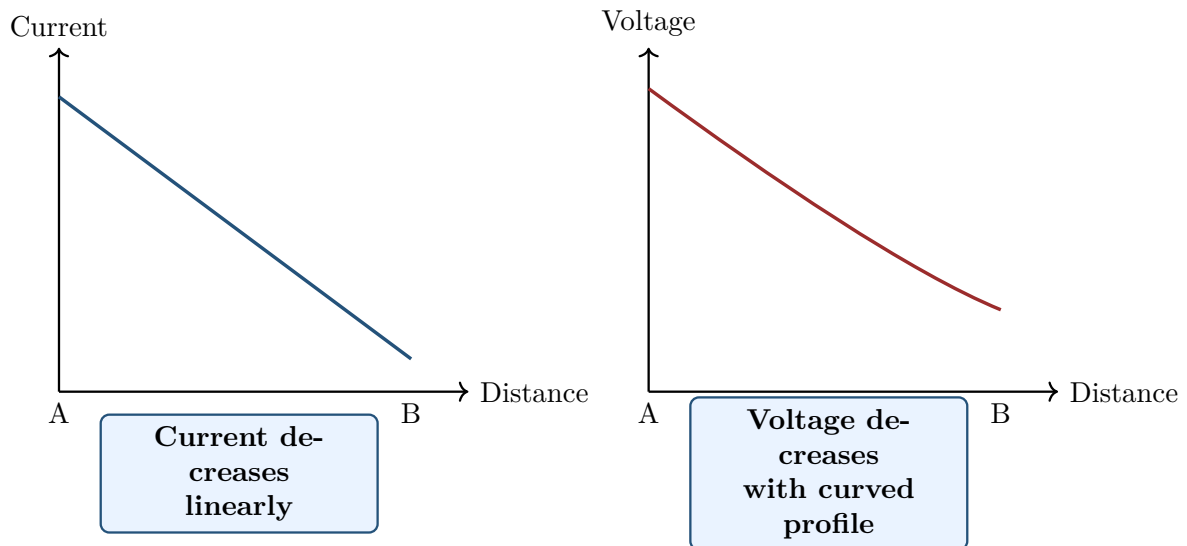
So current decreases linearly from il at A to zero at B.

13.2 Voltage Profile

Voltage drop from A to a point x is:

$$\Delta V_x = ir \left(lx - \frac{x^2}{2} \right)$$

So voltage does not fall as a straight line. It follows a curved profile.



How to Read the Diagram

Word/Symbol	Meaning
Current graph	Shows section current at different points of the uniformly loaded distributor.
Voltage graph	Shows voltage at different points along the distributor.
Distance	Horizontal axis measured from feeding end A to far end B.
Current	Current flowing in the distributor section at that distance.
Voltage	Voltage available at that point.
A	Feeding end. Current and voltage are maximum here.
B	Far end. Current becomes zero just beyond B and voltage is minimum at B.
Linear current fall	Because the remaining load beyond a point decreases uniformly.
Curved voltage fall	Because voltage drop depends on current, and current itself changes continuously.

14 Solved Example 2: Uniformly Loaded Distributor

14.1 Problem

A 2-wire DC distributor of length 500 m is fed at one end at 250 V. It has a uniformly distributed load of 0.4 A/m. Resistance of the distributor is $0.0006\Omega/\text{m}$ for both conductors. Find the voltage at the far end.

14.2 Solution

Given:

$$l = 500 \text{ m}$$

$$i = 0.4 \text{ A/m}$$

$$r = 0.0006 \Omega/\text{m}$$

Voltage drop for uniformly loaded distributor:

$$\Delta V = \frac{ir l^2}{2}$$

$$\Delta V = \frac{0.4 \times 0.0006 \times (500)^2}{2}$$

$$\Delta V = \frac{0.00024 \times 250000}{2}$$

$$\Delta V = 30 \text{ V}$$

Far-end voltage:

$$V_B = 250 - 30$$

$$V_B = 220 \text{ V}$$

Semester Answer Style

Since the distributor is fed at one end, the far end has the minimum voltage.

15 Case 3: Mixed Loading

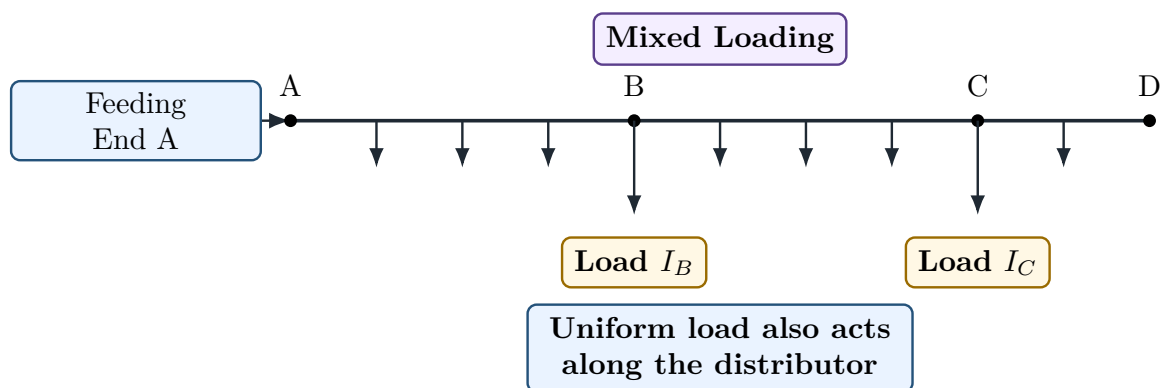
Definition

Mixed loading means the distributor has both concentrated loads and uniformly distributed load.

In mixed loading, solve in two parts:

Voltage drop due to concentrated loads + Voltage drop due to uniform load

15.1 Mixed Loading Diagram



How to Read the Diagram

Word/Symbol	Meaning
Mixed Loading	The distributor contains both point loads and distributed load.
Load I_B , Load I_C	Concentrated loads tapped at particular points B and C.
Uniform load	Load spread continuously along the length of the distributor.
Small downward arrows	Represent uniformly distributed load.
Long downward arrows	Represent concentrated loads at B and C.
A	Feeding end. Supply enters from here.
D	Far end. Usually this has the lowest voltage in one-end feeding.

16 Solved Example 3: Mixed Loading

16.1 Problem

A 2-wire DC distributor of length 400 m is fed at one end at 240 V. It has a uniformly distributed load of 0.2 A/m along the whole length. A concentrated load of 40 A is tapped at 200 m from the feeding end. Resistance of distributor is $0.0005\Omega/\text{m}$ for both conductors. Find far-end voltage.

16.2 Solution

Given:

$$l = 400 \text{ m}$$

$$i = 0.2 \text{ A/m}$$

$$I_c = 40 \text{ A at } 200 \text{ m}$$

$$r = 0.0005\Omega/\text{m}$$

16.3 Voltage Drop Due to Uniform Load

$$\Delta V_{\text{uniform}} = \frac{ir l^2}{2}$$

$$\Delta V_{\text{uniform}} = \frac{0.2 \times 0.0005 \times (400)^2}{2}$$

$$\Delta V_{\text{uniform}} = 8 \text{ V}$$

16.4 Voltage Drop Due to Concentrated Load

The 40 A load is at 200 m from feeding end. So the current 40 A flows only through first 200 m of distributor.

Resistance up to load point:

$$R = 200 \times 0.0005 = 0.1\Omega$$

Voltage drop due to concentrated load:

$$\Delta V_{\text{conc}} = 40 \times 0.1$$

$$\Delta V_{\text{conc}} = 4 \text{ V}$$

16.5 Total Voltage Drop

$$\Delta V_{\text{total}} = \Delta V_{\text{uniform}} + \Delta V_{\text{conc}}$$

$$\Delta V_{\text{total}} = 8 + 4 = 12 \text{ V}$$

Far-end voltage:

$$V_{\text{far}} = 240 - 12$$

$$V_{\text{far}} = 228 \text{ V}$$

17 General Table Method for Any One-End Distributor

For concentrated-load numericals, use this table format.

Section	Current in Section	Resistance	Voltage Drop
AB	Sum of all loads beyond A	R_{AB}	$I_{AB}R_{AB}$
BC	Loads beyond B	R_{BC}	$I_{BC}R_{BC}$
CD	Loads beyond C	R_{CD}	$I_{CD}R_{CD}$
DE	Loads beyond D	R_{DE}	$I_{DE}R_{DE}$

Semester Answer Style

This table is the safest method for semester numericals. It makes your solution clean, stepwise and easy to check.

18 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
Distributor fed at one end	Supply from one side only	Explain with diagram and voltage profile.
Current profile	Current decreases after every tapping	Draw current-loading diagram.
Voltage profile	Voltage falls along length	Draw voltage-drop diagram.
Concentrated loads	Loads at fixed points	Find voltage at each load point.
Uniform loading	Load spread along length	Derive $\Delta V = \frac{ir l^2}{2}$.
Mixed loading	Concentrated plus uniform load	Find far-end voltage using separate drops.
Minimum voltage	Lowest voltage point	For one-end feeding, it is normally at far end.

19 Most Predictable Semester Questions from Phase 4

No.	Question	Marks
1	Explain distributor fed at one end with diagram.	4
2	Draw current-loading and voltage profile diagram for distributor fed at one end.	4–5
3	A DC distributor fed at one end has concentrated loads at different points. Find voltage at each load point.	8–10
4	Derive voltage drop for a uniformly loaded distributor fed at one end.	5
5	Show that for uniformly loaded distributor fed at one end, voltage drop is $IR/2$.	4
6	Solve a mixed loading distributor problem with uniform and concentrated loads.	8
7	Explain why voltage is minimum at the far end of a distributor fed at one end.	3
8	Explain the importance of section current in distributor numericals.	3

20 Exam Answer Templates

20.1 Template 1: Theory Answer

Semester Answer Style

Answer sequence for “Distributor fed at one end”:

1. Give definition.
2. Draw clean distributor diagram.
3. Explain feeding end, load points and far end.
4. State that current is maximum near feeding end.
5. State that voltage is minimum at far end.
6. Draw current-loading and voltage profile if asked.

20.2 Template 2: Concentrated Load Numerical

Semester Answer Style

Answer sequence:

1. Draw the line diagram.
2. Mark loads and distances.
3. Find current in each section.
4. Find resistance of each section.
5. Find voltage drop of each section.
6. Subtract drops from supply voltage.
7. Box voltages at all load points.

20.3 Template 3: Uniform Load Derivation

Semester Answer Style

Answer sequence:

1. Draw uniformly loaded distributor.
2. Let load per metre be i , length l , resistance per metre r .
3. At distance x , current is $i(l - x)$.
4. Small voltage drop:

$$dV = i(l - x)r dx$$

5. Integrate from 0 to l .
6. Get:

$$\Delta V = \frac{ir l^2}{2} = \frac{IR}{2}$$

21 Phase 4 Final Memory Sheet

One-Page Memory Sheet

1. Distributor fed at one end receives supply from one side only.
2. Current is maximum near feeding end.
3. Current decreases after every load tapping.
4. Voltage is maximum at feeding end.
5. Voltage is minimum at far end.
6. Section current is sum of all loads beyond that section.
7. Voltage drop in a section is:

$$\Delta V = IR$$

8. Total voltage drop is sum of section drops.
9. Uniformly loaded distributor voltage drop:

$$\Delta V = \frac{ir l^2}{2}$$

10. If total current is I and total resistance is R :

$$\Delta V = \frac{IR}{2}$$

11. Mixed loading:

$$\Delta V_{\text{total}} = \Delta V_{\text{concentrated}} + \Delta V_{\text{uniform}}$$

22 Phase 4 Self-Test

Answer these without seeing notes:

1. What is a distributor fed at one end?
2. Why is current maximum near the feeding end?
3. Why is voltage minimum at the far end?
4. What is section current?
5. How do you find current in section AB?
6. How do you find voltage drop in a section?
7. What is the difference between concentrated and uniform load?
8. Derive voltage drop for a uniformly loaded distributor.
9. Why is $\Delta V = IR/2$ for uniformly loaded distributor?
10. How do you solve mixed loading problems?

Common Mistake

Do not jump directly to voltage calculation. In every distributor numerical:

Diagram → Section currents → Section drops → Voltages

End of Phase 4

Next Phase:

Phase 5 – Distributor Fed at Both Ends

Equal end voltages, unequal end voltages, minimum potential point, KVL method and semester numericals.